Chapter 1 Answer Key

Introduction to Statistics — MATH& 146

Section 1.1 Exercises

1.1.1. *Polling company calls 800 registered voters in Washington State.*

(a) **Population:** All registered voters in Washington State.

(b) **Sample:** The 800 registered voters who were called.

(c) **Parameter:** p, the true proportion of *all* registered voters in Washington State who approve of the governor.

(d) **Statistic:** $\hat{p}$, the proportion of the 800 sampled voters who approve (computed from the survey data).

1.1.2. *Veterinarian measures 30 cats; sample mean weight is 10.4 lbs.*

(a) **Population:** All domestic cats in the veterinarian's city.

(b) **Sample:** The 30 cats brought in for annual checkups.

(c) **Parameter:** μ, the true mean weight of all domestic cats in the city (unknown).

(d) **Statistic:** $\bar{x} = 10.4$ lbs, the mean weight of the 30 sampled cats.

1.1.3. **False.** A parameter is a fixed characteristic of the *population*, not of any particular sample. It does not change when a new sample is taken. Statistics vary from sample to sample; parameters do not. (We may not know the parameter's value, but it is not moving around — our estimates of it are.)

Section 1.2 Exercises

1.2.1. Classifications:

(a) Number of cars in a household: **discrete quantitative** (countable whole numbers).

(b) Annual income: **continuous quantitative** (can take any non-negative value along a scale).

(c) Brand of coffee preferred: **categorical**.

(d) Blood type: **categorical**.

(e) Distance driven to campus: **continuous quantitative**.

(f) Score on a 10-question quiz: **discrete quantitative** (whole numbers 0 through 10).

1.2.2. Answers will vary. Example: *Categorical* — declared major. *Continuous quantitative* — height in inches.

Section 1.3 Exercises

Given $x_1 = 4,\ x_2 = 1,\ x_3 = 6,\ x_4 = 3.$

1.3.1. (a) $\sum_{i=1}^{4} x_i = 4 + 1 + 6 + 3 = \mathbf{14}$

(b) $\sum_{i=1}^{4} x_i^2 = 4^2 + 1^2 + 6^2 + 3^2 = 16 + 1 + 36 + 9 = \mathbf{62}$

(c) $\left(\sum_{i=1}^{4} x_i\right)^2 = 14^2 = \mathbf{196}$

Note that $62 \neq 196$, illustrating the pitfall discussed in the text.

1.3.2. Using the linearity of summation:

$$\sum_{i=1}^{6}(x_i + 3) = \sum_{i=1}^{6} x_i + \sum_{i=1}^{6} 3 = 42 + 3(6) = 42 + 18 = \mathbf{60}.$$

1.3.3. In plain English: "For each data value x_i, subtract the mean $\bar{x}$ from it to get its deviation; then add up all n of those deviations." The result is always zero — the positive and negative deviations cancel exactly. This is why we cannot use the sum of deviations as a measure of spread.

Section 1.4 Exercises

1.4.1. Fifteen quiz scores: 70, 82, 91, 75, 88, 70, 95, 82, 77, 91, 70, 88, 82, 75, 91.

Score	Frequency	Relative Frequency
70	3	$3/15 = 0.200$
75	2	$2/15 \approx 0.133$
77	1	$1/15 \approx 0.067$
82	3	$3/15 = 0.200$
88	2	$2/15 \approx 0.133$
91	3	$3/15 = 0.200$
95	1	$1/15 \approx 0.067$
Total	15	1.000

1.4.2. Scores 70, 82, and 91 are *tied* for the highest relative frequency, each appearing 3 times. Each accounts for $3/15 = 0.20$, or **20%** of the class.

1.4.3. When two groups have different sizes (15 vs. 30 students), raw frequency counts are not directly comparable: a score appearing 6 times in the larger class is no more "common" than a score appearing 3 times in the smaller class. Relative frequencies standardize for group size, expressing each count as a fraction of its own total, making fair comparisons possible.

Section 1.5 Exercises

1.5.1. Data: $14, 22, 9, 31, 17, 22, 8$. $\quad n = 7$.

Mean:

$$\bar{x} = \frac{14 + 22 + 9 + 31 + 17 + 22 + 8}{7} = \frac{123}{7} \approx \mathbf{17.57}.$$

Median: Sorted data: $8, 9, 14, \mathbf{17}, 22, 22, 31$. With $n = 7$ (odd), the median is at position $(7 + 1)/2 = 4$. The 4th value is **17**.

1.5.2. Adding an observation of 200 to a dataset whose current mean is 50: since $200 > 50$, the new value lies well above the current mean and will **increase** it. How much depends on the original sample size n: if n is large, the effect is small; if n is small, the increase can be substantial.

1.5.3. House prices: \$180k, \$195k, \$205k, \$212k, \$950k.

Mean:

$$\bar{x} = \frac{180 + 195 + 205 + 212 + 950}{5} = \frac{1742}{5} = \$348.4\text{k}.$$

Median: Already sorted; $n = 5$, so the median is the 3rd value: \$**205**k.

The **median** is the better summary. The \$950k value is a high outlier that pulls the mean up to \$348.4k — far above four of the five prices. The median of \$205k accurately reflects the center of the distribution for a typical home in this neighborhood.

Section 1.6 Exercises

1.6.1. Data: $10, 14, 12, 16, 13$. $\quad n = 5$. $\quad \bar{x} = (10 + 14 + 12 + 16 + 13)/5 = 65/5 = 13$.

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
10	-3	9
14	$+1$	1
12	-1	1
16	$+3$	9
13	0	0
Sum	0	20

$$s = \sqrt{\frac{20}{5 - 1}} = \sqrt{\frac{20}{4}} = \sqrt{5} \approx \mathbf{2.24}.$$

1.6.2. Both classes have a mean of 75, so their average performance is the same. However, Class 1 ($s = 3$) is far more consistent: nearly every student scored close to 75, with very little variation. Class 2 ($s = 18$) shows much greater spread — scores ranged widely above and below 75, indicating substantial differences in individual performance.

1.6.3. **Dataset B** has the larger standard deviation. Dataset A contains values 99, 100, 101, 100, 100 — all within 1 unit of the mean of 100. Dataset B contains values 80, 90, 100, 110, 120, which deviate from their mean of 100 by as much as 20 units. Greater distance from the mean means greater standard deviation.

Section 1.7 Exercises

1.7.1. The distribution would be **skewed right**. Most ER patients are younger adults, so data cluster at lower ages. A smaller number of very elderly patients create a long tail extending toward higher ages. Since the tail points right, the distribution is right-skewed.

1.7.2. With mean $= 45 <$ median $= 52$, the mean is pulled *below* the median. This indicates a **left-skewed** (negatively skewed) distribution, with a long tail extending toward lower values.

Chapter 1 Review

1. Researcher surveys 200 of 5,000 employees for average commute time.

(a) **Population:** All 5,000 employees at the company.

(b) **Sample:** The 200 employees who were surveyed.

(c) **Parameter:** μ, the true mean commute time for all 5,000 employees (unknown).

(d) **Statistic:** $\bar{x}$, the mean commute time computed from the 200 surveyed employees.

2. Ages: 18, 19, 21, 20, 22, 19, 18, 25, 20, 21, 18, 19, 30, 20, 21. $n = 15$.

(a) **Frequency and relative frequency table:**

Age	Frequency	Relative Frequency
18	3	$3/15 = 0.200$
19	3	$3/15 = 0.200$
20	3	$3/15 = 0.200$
21	3	$3/15 = 0.200$
22	1	$1/15 \approx 0.067$
25	1	$1/15 \approx 0.067$
30	1	$1/15 \approx 0.067$
Total	15	1.000

(b) **Mean and median:**

$$\bar{x} = \frac{311}{15} \approx 20.73.$$

Sorted data: 18, 18, 18, 19, 19, 19, 20, **20**, 20, 21, 21, 21, 22, 25, 30. With $n = 15$, the median is at position 8: median = **20**.

The mean (20.73) and median (20) are close. Ages 18–21 dominate, so the center is well-defined. The small gap reflects a mild right skew from the three older students (22, 25, 30).

(c) **Standard deviation.**
Using $\bar{x} = 311/15$:

Age	Frequency	$x_i - \bar{x}$	$(x_i - \bar{x})^2 \times$ freq
18	3	$-41/15 \approx -2.733$	$3(1681/225) = 5043/225$
19	3	$-26/15 \approx -1.733$	$3(676/225) = 2028/225$
20	3	$-11/15 \approx -0.733$	$3(121/225) = 363/225$
21	3	$4/15 \approx 0.267$	$3(16/225) = 48/225$
22	1	$19/15 \approx 1.267$	$1(361/225) = 361/225$
25	1	$64/15 \approx 4.267$	$1(4096/225) = 4096/225$
30	1	$139/15 \approx 9.267$	$1(19321/225) = 19321/225$
		Sum $= 0$	$31260/225 \approx 138.93$

$$s = \sqrt{\frac{138.93}{14}} = \sqrt{9.924} \approx \mathbf{3.15}.$$

(d) **Shape:** The distribution is mildly **skewed right**. The bulk of students are 18–21, but the three values at 22, 25, and 30 extend a tail to the right, pulling the mean slightly above the median.

3. Given $x_1 = 2, x_2 = 5, x_3 = 3, x_4 = 8, x_5 = 2$.

(a) $\sum x_i = 2 + 5 + 3 + 8 + 2 = \mathbf{20}$

(b) $\sum x_i^2 = 4 + 25 + 9 + 64 + 4 = \mathbf{106}$

(c) $\left(\sum x_i\right)^2 = 20^2 = \mathbf{400}$

(d) $\bar{x} = \dfrac{20}{5} = \mathbf{4}$

(e) Deviation table with $\bar{x} = 4$:

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
2	-2	4
5	$+1$	1
3	-1	1
8	$+4$	16
2	-2	4
Sum	0	26

$$s = \sqrt{\frac{26}{4}} = \sqrt{6.5} \approx \mathbf{2.55}.$$

4. Model response (writing prompt):

The standard deviation measures the typical distance between individual data values and their mean: a small standard deviation indicates that observations cluster tightly around the mean, while a large one indicates they are widely scattered. We divide by $n - 1$ rather than n because we are estimating the population standard deviation from a sample; using $\bar{x}$ as a stand-in for the unknown μ consumes one degree of freedom, and dividing by $n - 1$ corrects for the resulting tendency to underestimate the true spread.